EXP NO : 9 Implementation of Subnetting in CISCO Packet tracer simulator.

AIM:

TO design multiple subnets, assign static, IP address connects the subnet using a router, and simulator packet transmission in cisco packet tracer.

objective :

*) TO create multiple subnet with a suitable number of hosts.

*) TO assign static IP addresses to all devices

*) TO connect different subnet using a router

*) TO test communication between subnet with and without a router.

Requirements :

*) cisco packet tracer

*) 1 router

*) 2 switches

*) 4 PCS

a) Design multiple subnet with suitable number of host creates two subnet using cisco packet tracer with 1 router, a switche and 4 pcs.

* connect two pcs to each switch and connect both switche to the router.

*) Subnet 1; 192.168.1.0/26

*) Subnet 2; 192.168.1.64/26

b) Assign static IP address across all subnet and connect the subnet via router.

b) Assign static IP address across all subnet and connect the subnet via router,

Device	IP address	Subnet mask
PC0	192.168.1.2	255.255.255.192
PC1	192.168.1.3	255.255.255.192
PC2	192.168.1.66	255.255.255.192
PC3	192.168.1.67	255.255.255.192
Rauter Gr0/0	192.168.1.1	255.255.255.192
Router Gr0/1	192.168.1.65	255.255.255.192

c) Simulate packet transmission across the subnet and observe the result

using the ping command to test communication between PCs in different Subnet.

Ping 192.168.1.66

a) when Subnet are connected via a router

Packet are successfully transmitted between the two Subnet the router forward packet to the destination network, and the ping command receive replies.

b) when Subnet are not connected without a router

PCs in the same Subnet can communicate through the ~~Subnet~~ Switch. Pcs in different Subnet cannot communicate because there is no router to forward packet between them,

Result: multiple subnet were designed in cisco packet tracer and static IP address were assigned to all device. communication between Subnet was tested with and without a router.